Self-similar counterexamples to three conjectures of Lorentz

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Abstract

We give constructions for three related symmetry conjectures in uniform polynomial approximation. One continuous nonodd function has every best approximating polynomial vanish at the origin. A second continuous nonodd function, normalized to vanish at zero, has equal best approximants at every pair of degrees $(2k+1, 2k+2)$. A third continuous function is not even but has equal best approximants at every pair $(2k, 2k+1)$. Each construction is a uniformly convergent Chebyshev series with polynomial forcing. Exact finite algebraic certificates, analytic persistence estimates, and cubic self-similarity prove the assertions at all degrees. The finite computations use rational arithmetic, and the manuscript records the separate status of the Lean formalizations.

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1 Common notation and the composition principle

Let $I = [-1, 1]$, let $P_n f$ denote the unique best uniform approximant to $f \in C(I)$ among real polynomials of degree at most n, and let $T_k(\cos\theta) = \cos(k\theta)$. We consider the conjectures in the formulation of Saff and Varga [1]. The conclusions concern, respectively,

$$(P_n f)(0) = 0 \ (n \geq 0), \qquad P_{2k+1} f = P_{2k+2} f \ (k \geq 0), \qquad P_{2k} f = P_{2k+1} f \ (k \geq 0).$$

The two parts below use different contraction parameters and different forcing polynomials. Their scalar notation is local to each part; the operator P_n and the polynomials T_k retain the above meaning throughout.

Lemma 1.1 (Chebyshev composition)**.** *For $F \in C(I)$, integers $m \geq 1$ and $n \geq 0$,*

$$P_n(F \circ T_m) = (P_{\lfloor n/m \rfloor} F) \circ T_m.$$

Proof. Average any competitor $p(\cos\theta)$ over $\theta + 2\pi j/m$, $0 \leq j < m$. The target $F(\cos m\theta)$ is unchanged, so this average has no larger uniform error. Only cosine frequencies multiple of m remain, giving $r(\cos m\theta)$ with $\deg r \leq \lfloor n/m \rfloor$. Surjectivity of $T_m : I \to I$ gives equality of the two best errors. Uniqueness of best approximation gives the identity. □

Part I
Every best approximant vanishes at the origin

2 Construction and reduction to seven degrees

Write $I = [-1, 1]$, let $T_k(\cos\theta) = \cos(k\theta)$, and let $P_n f$ denote the unique best uniform approximant to $f \in C(I)$ among polynomials of degree at most n. We address the origin interpolation implication discussed in [1]. Set $z = 2\cos(\pi/36)$ and

$$\begin{aligned} 107\beta = \ & -60 + 82z + 1086z^2 + 1576z^3 - 2932z^4 - 3433z^5 \\ & + 2002z^6 + 2243z^7 - 510z^8 - 587z^9 + 44z^{10} + 54z^{11}. \end{aligned}$$

In particular $419/1000 < \beta < 420/1000$. Define

$$h = T_4 - 1 - 2T_5 + \beta T_7, \qquad q_\varepsilon = T_3 + \varepsilon h, \qquad f_\varepsilon = \sum_{j \geq 0} (-1/3)^j q_\varepsilon \circ T_{3^j}.$$

The series converges uniformly on I, so f_ε is continuous.

Theorem 2.1. *For every $0 < \varepsilon \leq 1/25000000$, the above function satisfies*

$$(P_n f_\varepsilon)(0) = 0 \quad (n \geq 0), \qquad f_\varepsilon(1/\sqrt{2}) + f_\varepsilon(-1/\sqrt{2}) = -3\varepsilon.$$

Thus it is not odd.

By Lemma 1.1, the construction reduces to seven degrees.

The functional equation and its consequence for $n \geq 7$ are

$$f_\varepsilon = q_\varepsilon - \tfrac{1}{3} f_\varepsilon \circ T_3, \qquad P_n f_\varepsilon = q_\varepsilon - \tfrac{1}{3}(P_{\lfloor n/3 \rfloor} f_\varepsilon) \circ T_3. \tag{1}$$

Here we used polynomial translation, scalar homogeneity, and $\deg q_\varepsilon \leq 7$. Since $q_\varepsilon(0) = T_3(0) = 0$, induction reduces the origin condition to $0 \leq n \leq 6$.

3 Baseline residuals and the finite certificate

Put

$$a = 1/\sqrt{2}, \quad G = \sum_{j \geq 0} (-1/3)^j T_{3^j}, \quad F = G \circ T_3, \quad H = \sum_{j \geq 0} (-1/3)^j h \circ T_{3^j}.$$

Then $f_\varepsilon = F + \varepsilon H$. Pairing consecutive terms gives

$$G = \sum_{j \geq 0} 9^{-j} g \circ T_{9^j}, \qquad g(x) = 2x - \tfrac{4}{3} x^3.$$

The unique maximum of g on I is at a, its unique minimum is at $-a$, and $T_9(\pm a) = \pm a$. Hence G has the same unique extrema and

$$E := \|G\|_\infty = \frac{g(a)}{1 - 1/9} = \frac{3a}{2}.$$

Oddness gives $P_0 G = 0$. From $G = x - G \circ T_3/3$ and Lemma 1.1, $P_1 G = P_2 G = x$. Thus for the seven base degrees,

$$p_n^0 = P_n F = \begin{cases} 0 & n \leq 2, \\ T_3 & 3 \leq n \leq 6, \end{cases} \qquad b_n := F - p_n^0 = \begin{cases} G \circ T_3 & n \leq 2, \\ -G \circ T_9/3 & 3 \leq n \leq 6. \end{cases} \tag{2}$$

Their norms E_n are respectively E and $E/3$. Their contact sets S_n are respectively the zeros of T_6 and T_{18}.

Here are exact definitions of the finite certificate. Use the increasing nodes

$$X_j = -\cos \frac{(2j-1)\pi}{36} \quad (1 \leq j \leq 18), \qquad (C_1, \ldots, C_6) = (X_2, X_5, X_8, X_{11}, X_{14}, X_{17}).$$

The residual signs on the fine grid repeat $(+, -, -, +)$; call them σ_j. On the coarse grid they repeat $(-, +, +, -)$; call them τ_j. The two-cycle $T_3(a) = -a$, $T_3(-a) = a$ gives

$$H(\pm a) = -\tfrac{3}{2} \pm \tfrac{3}{2}(2 + \beta) a.$$

Iterating $H = h - H \circ T_3/3$ twice, and using $T_{21} = -T_{15}$ at the zeros of T_{18}, yields $H(X_j) = R_\beta(X_j)$, where

$$R_\beta = T_4 - \tfrac{1}{3} T_{12} - \tfrac{5}{6} - 2T_5 + \beta T_7 + \tfrac{2+\beta}{3} T_{15} + \tfrac{2+\beta}{6} T_9. \tag{3}$$

Define corrections v_n and slopes λ_n as follows:

$$\begin{aligned} v_0 &= 0, & \lambda_0 &= \tfrac{\sqrt{6}}{4}(2 + \beta), \\ v_1 = v_2 &= -(2 + \beta) x, & \lambda_1 = \lambda_2 &= \tfrac{2+\beta}{2\sqrt{2}}, \\ v_3 &= A_1 x + A_2 x^2 + A_3 x^3, \\ v_4 &= T_4 - 1 + B_1 T_1 + B_3 T_3, \\ v_5 = v_6 &= T_4 - 1 + D_1 T_1 + D_3 T_3 + D_5 T_5, & \lambda_6 &= \lambda_5. \end{aligned}$$

The remaining coefficients and slopes are the solutions of

$$v_n(X_j) + \sigma_j \lambda_n = R_\beta(X_j), \qquad \begin{array}{c|l} n & j \\ \hline 3 & 1,3,8,10 \\ 4 & 10,16,18 \\ 5 & 10,13,15,16. \end{array} \tag{4}$$

The three defining matrices are nonsingular, as certified below. In particular $\deg v_n \le n$ and $v_n(0) = 0$.

For a contact $\xi \in S_n$, write $s(\xi) = \operatorname{sgn} b_n(\xi)$. The exact finite calculation establishes

$$s(\xi)\big(H(\xi) - v_n(\xi)\big) \le \lambda_n, \qquad 0 < \lambda_n < 2, \tag{5}$$

with the following complete zero-slack sets. Indices refer to the indicated grid.

n	grid	zero-slack indices	other slacks exceed
0	C	$3,4$	$127/50$
1,2	C	$1,3,4,6$	$363/100$
3	X	$1,3,8,10,16$	$9/100$
4	X	$1,3,8,9,10,16,18$	$7/10$
5,6	X	$3,4,6,9,10,13,15,16$	$1/5$

For $n = 0, 2, 3, 6$, use the whole listed set. For $n = 1, 4, 5$ use respectively $\{1,3,4\}$, $\{1,3,9,10,16,18\}$, and $\{3,4,6,9,10,13,15\}$. Each selected set contains $n+2$ increasing nodes with alternating signs.

The sums of the absolute Chebyshev coefficients of v_n are respectively

$$0, \quad < 5/2, \quad < 5/2, \quad < 9/4, \quad < 13/4, \quad < 11/2, \quad < 11/2.$$

Consequently $\|v_n\|_\infty < 6$ and $\|v_n'\|_\infty < 216$, since $\|T_k'\|_\infty = k^2$ and $\deg v_n \le 6$.

Exact verification of the finite assertions

All the numbers above lie in $\mathbb{Q}[z]$, with arithmetic reduced modulo

$$m(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1 = (z^6 - 6z^4 + 9z^2 - 2)^2 - 3.$$

The last identity and $z = 2\cos(\pi/36)$ give $m(z) = 0$. There is a unique root in $(1.98, 2)$: the inner polynomial is positive there and has derivative $6z(z^2-1)(z^2-3) > 0$. Rational endpoint evaluations place the root in

$$[199238939/10^8,\ 199238940/10^8].$$

The accompanying `scripts/check_certificate.py` reads exact rational coefficient lists from `CertificateData.lean` and `CertificateGrid.lean`. It reconstructs the nodes by the Chebyshev recurrence, verifies (3), the functional equation on the grid, every slack identity, every equality set, alternation, coefficient norms, and the nonzero determinants in (4). Equality is tested as equality of reduced rational polynomials. One hundred rational bisections, followed by rational interval Horner evaluation, certify the strict inequalities. No floating-point tolerance is used. This also verifies that the stored coefficients are the uniquely defined solutions of (4).

4 Signed cusps and a global bound relative to contacts

The two estimates needed to turn (5) into uniform approximation are proved here with explicit constants.

Lemma 4.1. *For $s \in \{-1, 1\}$ set $S_{n,s} = \{\xi \in I : b_n(\xi) = sE_n\}$. For $0 \le n \le 6$ and $x \in I$,*

$$E_n - sb_n(x) \ge \frac{1}{250} \operatorname{dist}(x, S_{n,s}).$$

Proof. First we prove $E - sG(y) \ge c_G|y - sa|$ on I, where $c_G = \pi(\sqrt{2} - 1)/324$. The factorization

$$g(a) - g(y) = \tfrac{4}{3}(y - a)^2(y + \sqrt{2}) \ge C_0(y - a)^2, \qquad C_0 = \tfrac{4}{3}(\sqrt{2} - 1),$$

makes every term of $E - G(y) = \sum_{j \ge 0} 9^{-j}(g(a) - g(T_{9^j}(y)))$ nonnegative. Put $\delta = \pi/12$ and write $y = \cos(\pi/4 + u)$. For $0 < |u| \le \delta$, choose the largest $j \ge 0$ with $9^j|u| \le \delta$, so $\delta/9 < 9^j|u| \le \delta$. Since $9^j \equiv 1 \pmod 8$ and $\sin t \ge 1/2$ for $t \in [\pi/6, \pi/3]$, this single summand gives

$$E - G(y) \ge \frac{C_0}{4} 9^{-j}(9^j u)^2 > \frac{C_0 \delta}{36}|u| \ge c_G|y - a|.$$

For $|u| > \delta$, we have $|y - a| \ge r_0 = (\sqrt{3} - \sqrt{2})/2$. The first summand then gives the result because $C_0 r_0 > c_G$. At $u = 0$ it is immediate. Oddness of G proves the negative-sign case.

For every integer $L \ge 1$ and $s = \pm 1$ we also have the signed pullback bound

$$|T_L(x) - sa| \ge \kappa L \operatorname{dist}(x, \{\xi : T_L(\xi) = sa\}), \qquad \kappa = 4(1 - a)/\pi. \tag{6}$$

To see this, on $[0, \pi]$ the inequality $|\cos t - a| \ge \kappa|t - \pi/4|$ follows from the concavity chord on $[0, \pi/4]$ and $\sin t \ge a > \kappa$ on $[\pi/4, \pi/2]$. On $[\pi/2, \pi]$, convexity gives $\cos t \le 1 - 2t/\pi$, and

$$a - \cos t \ge 2t/\pi - (1 - a) \ge 4(1 - a)(t - \pi/4)/\pi,$$

because $a \ge 1/2$. Reflection gives the analogous assertion for $-a$. Apply these inequalities on the half-period of $L\theta$ containing the given angle, where $x = \cos\theta$. The unique preimage of the prescribed target in that half-period lies in $[0, \pi]$ after division by L. The inequality $|\cos\theta - \cos\phi| \le |\theta - \phi|$ proves (6).

For $n \le 2$ use $L = 3$ and target sa; for $n \ge 3$ use $L = 9$ and target $-sa$, with the factor $1/3$ in (2). Both give

$$E_n - sb_n(x) \ge 3c_G\kappa \operatorname{dist}(x, S_{n,s}), \qquad 3c_G\kappa = \frac{3 - 2\sqrt{2}}{27\sqrt{2}} > \frac{1}{250}.$$

The final strict inequality is equivalent to $750 > 527\sqrt{2}$, verified by squaring positive quantities. □

Lemma 4.2. *For $r_n = H - v_n$, every $0 \le n \le 6$, every contact $\xi \in S_n$, and every $x \in I$,*

$$|r_n(x) - r_n(\xi)| \le 10^5|x - \xi|.$$

This is a bound relative to each fixed contact; it asserts no global Lipschitz regularity of H.

Proof. Pair the perturbation series:

$$H = \sum_{j\ge 0} 9^{-j} Q \circ T_{9^j}, \qquad Q = h - h \circ T_3/3.$$

Since $T_3'(\pm a) = 3$ and $h_{\text{even}} = T_4 - 1$, we have $Q'(\pm a) = 0$. For $x = \cos(\pi/4 + u)$, $0 < |u| \le \delta = \pi/12$, choose the last scale j as in Lemma 4.1. Taylor's theorem and a geometric tail bound give

$$\begin{aligned} |H(x) - H(a)| &\le \tfrac{1}{2} \left\|Q''\right\|_\infty u^2 \sum_{k=0}^{j} 9^k + 2 \left\|Q\right\|_\infty \sum_{k=j+1}^{\infty} 9^{-k} \\ &\le \left(\tfrac{9\delta}{16} \left\|Q''\right\|_\infty + \tfrac{9}{4\delta} \left\|Q\right\|_\infty\right) |u|. \end{aligned}$$

Here $|x - a| \ge |u|/2$. The same proof around $3\pi/4$ applies to $-a$. Thus, whenever $|x \mp a| \le r_0$, a valid constant is

$$L_H = \tfrac{3\pi}{32} \left\|Q''\right\|_\infty + \tfrac{54}{\pi} \left\|Q\right\|_\infty < 7000.$$

For the numerical bound, $\beta < 0.420$ and the Chebyshev derivative formulas $\left\|T_k'\right\|_\infty = k^2$, $\left\|T_k''\right\|_\infty = k^2(k^2-1)/3$ give

$$\left\|h\right\|_\infty < 4.42, \qquad \left\|h'\right\|_\infty < 87, \qquad \left\|h''\right\|_\infty < 810,$$

and hence $\left\|Q\right\|_\infty < 5.894$ and $\left\|Q''\right\|_\infty \le 28 \left\|h''\right\|_\infty + 8 \left\|h'\right\|_\infty < 23376$. Using $3 < \pi < 22/7$ yields the displayed bound for L_H.

If $L = 3^k$ and $T_L(\xi) = \pm a$, iterate the functional equation k times. The derivative bound for its finite prefix is $\left\|h'\right\|_\infty \sum_{i=0}^{k-1} 3^i = (L-1) \left\|h'\right\|_\infty /2$. Since $\left\|T_L'\right\|_\infty = L^2$, this gives

$$|H(x) - H(\xi)| \le \left(\tfrac{L-1}{2} \left\|h'\right\|_\infty + L L_H\right) |x - \xi| \quad \text{if } |x - \xi| \le r_0/L^2.$$

For our two grids $L = 3, 9$, and $r_0/81 > 10^{-3}$. Adding the derivative bound on v_n gives a constant less than $4 \cdot 87 + 9 \cdot 7000 + 216 < 10^5$ when $|x - \xi| \le 10^{-3}$. The inequality is non-strict, including $x = \xi$.

Finally $\left\|H\right\|_\infty \le \frac{3}{2} \left\|h\right\|_\infty < 6.63$, so $\left\|r_n\right\|_\infty < 20$. For $|x - \xi| > 10^{-3}$, $|r_n(x) - r_n(\xi)| < 40 < 10^5 |x - \xi|$, which extends the estimate to all I. □

5 One persistence inequality and completion

Fix a base degree n and a sign $s \in \{-1, 1\}$. Choose a nearest contact $\xi \in S_{n,s}$ and put $d = |x - \xi|$. Lemmas 4.1 and 4.2, together with (5), give

$$\begin{aligned} s\big(b_n(x) + \varepsilon r_n(x)\big) &\le E_n - d/250 + \varepsilon\big(\lambda_n + 10^5 d\big) \\ &= E_n + \varepsilon\lambda_n - (1/250 - 10^5\varepsilon) d \\ &\le E_n + \varepsilon\lambda_n \qquad \left(0 \le \varepsilon \le \frac{1}{25000000}\right). \end{aligned}$$

Apply both signs. This proves

$$\left\|f_\varepsilon - (p_n^0 + \varepsilon v_n)\right\|_\infty \le E_n + \varepsilon\lambda_n.$$

At the $n+2$ alternating equality nodes the residual is exactly $s(\xi)(E_n + \varepsilon\lambda_n)$. Its magnitude is positive, since $E_n > 0$ and $\lambda_n > 0$. The alternation theorem identifies $P_n f_\varepsilon = p_n^0 + \varepsilon v_n$. All these polynomials vanish at zero, and (1) proves the assertion for every degree.

For nonoddness, F is odd and T_3 interchanges a and $-a$. At both points $T_4 - 1 = -2$, so

$$H(a) + H(-a) = \sum_{j\geq 0}(-1/3)^j(-4) = -3.$$

It follows that $f_\varepsilon(a) + f_\varepsilon(-a) = -3\varepsilon \neq 0$, completing Theorem 2.1.

Part II
Consecutive-degree plateaus without the predicted symmetry

6 The statements and the proof strategy

For $f \in C[-1,1]$, write $P_n f$ for its unique best uniform approximant among real polynomials of degree at most n, and let $E_n(f) = \|f - P_n f\|_\infty$. Throughout, T_k is the Chebyshev polynomial defined by $T_k(\cos\theta) = \cos(k\theta)$.

The two assertions considered here are the normalized oddness conjecture and the evenness conjecture from [1]:

$$\begin{aligned} f(0) = 0, \quad P_{2k+1}f = P_{2k+2}f \ (k \geq 0) \quad &\Longrightarrow \quad f \text{ is odd}, \\ P_{2k}f = P_{2k+1}f \ (k \geq 0) \quad &\Longrightarrow \quad f \text{ is even}. \end{aligned}$$

Theorem 6.1. *There exist $f_{\mathrm{o}}, f_{\mathrm{e}} \in C[-1,1]$ such that*

$$\begin{aligned} f_{\mathrm{o}}(0) = 0, \qquad P_{2k+1}f_{\mathrm{o}} = P_{2k+2}f_{\mathrm{o}} \quad (k \geq 0), \qquad & f_{\mathrm{o}}(1) + f_{\mathrm{o}}(-1) \neq 0, \\ P_{2k}f_{\mathrm{e}} = P_{2k+1}f_{\mathrm{e}} \quad (k \geq 0), \qquad & f_{\mathrm{e}}(1) - f_{\mathrm{e}}(-1) \neq 0. \end{aligned}$$

Both functions are specified below by algebraic polynomial forcing and the fixed parameter $\varepsilon = 10^{-10}$.

The proof has three dependencies. First, Chebyshev composition reduces all degree conditions to finitely many initial ones. Second, an explicit baseline has known extremal points and sufficiently strong local gaps to preserve finite contact inequalities under perturbation. Third, exact algebraic certificates supply the required initial approximants. We prove these ingredients in that order and then assemble the two constructions.

7 Self-similarity and the finite degree reduction

Set

$$a = \frac{3}{5}, \qquad u = \frac{\sqrt{2}}{2}, \qquad \mathcal{S}h(x) = \sum_{j=0}^{\infty}(-a)^j h(T_{3^j}(x)), \qquad G = \mathcal{S}T_1.$$

For continuous h, the series converges uniformly on $[-1,1]$, since $\|h \circ T_{3^j}\|_\infty = \|h\|_\infty$ and $\sum a^j < \infty$. In particular, $\mathcal{S}h$ is continuous and

$$\mathcal{S}h = h - a(\mathcal{S}h) \circ T_3. \tag{7}$$

Commutation of Chebyshev polynomials gives $\mathcal{S}T_d = G \circ T_d$.

Lemma 7.1 (Recurrence for best approximants)**.** *If q is a polynomial and $f = \mathcal{S}q$, then*

$$P_n f = q - a(P_{\lfloor n/3 \rfloor} f) \circ T_3, \qquad n \geq \deg q. \tag{8}$$

Proof. For $n \geq \deg q$, translation by q preserves the space of degree-at-most n polynomials. Apply this translation, homogeneity of best approximation, (7), and Lemma 1.1. □

Proposition 7.2 (Finite conditions for the two parity patterns)**.** *Let $f = \mathcal{S}q$.*

1. *If $\deg q \leq 7$ and $P_1 f = P_2 f$, $P_3 f = P_4 f$, and $P_5 f = P_6 f$, then $P_{2k+1} f = P_{2k+2} f$ for every $k \geq 0$.*
2. *If $\deg q \leq 8$ and $P_0 f = P_1 f$, $P_2 f = P_3 f$, $P_4 f = P_5 f$, and $P_6 f = P_7 f$, then $P_{2k} f = P_{2k+1} f$ for every $k \geq 0$.*

Proof. For the first claim, use strong induction on odd n. The cases $n = 1, 3, 5$ are assumptions. For odd $n \geq 7$, the two floor indices in (8) for $n, n+1$ agree unless $n = 6r + 5$. In that case they are $2r + 1, 2r + 2$, a smaller odd/even pair covered by induction.

For the second claim, use strong induction on even n. The cases $n = 0, 2, 4, 6$ are assumptions. For even $n \geq 8$, the floor indices agree unless $n = 6r + 2$, in which case they are the smaller even/odd pair $2r, 2r + 1$. The same recurrence completes the proof. □

8 The baseline and its exact extrema

For $0 < c < 1$, define

$$G_c(x) = \sum_{j \geq 0} (-c)^j T_{3^j}(x).$$

Thus $G = G_a$. Every G_c is odd. We first establish the extremal geometry for $b = 1/3$ and then transfer it to $a = 3/5$.

Lemma 8.1. *The function G has unique maximum at u, unique minimum at $-u$, and*

$$B := G(u) = \|G\|_\infty = \frac{u}{1 - a} = \frac{5u}{2}. \tag{9}$$

Moreover, for every $y \in [-1, 1]$,

$$B - |G(y)| \geq \frac{1}{2} \operatorname{dist}(y, \{-u, u\})^2. \tag{10}$$

Proof. Pair adjacent terms in G_b:

$$G_b = \sum_{j \geq 0} 9^{-j} g \circ T_{9^j}, \qquad g(x) = x - \frac{1}{3} T_3(x) = 2x - \frac{4}{3} x^3.$$

The identity

$$g(u) - g(y) = \frac{4}{3}(y - u)^2 (y + 2u)$$

shows that g has its unique maximum at u and its unique minimum at $-u$ on the interval. Since $T_9(u) = u$, every term in the paired series has its maximum at u. Consequently $B_b := \|G_b\|_\infty = 3u/2$ and

$$B_b - G_b(y) \geq \frac{4}{3}(y - u)^2 (y + 2u) \geq \frac{1}{2}(y - u)^2. \tag{11}$$

The last inequality uses $\frac{4}{3}(\sqrt{2}-1) > 1/2$. Oddness gives the corresponding lower bound for $B_b + G_b(y)$.

The uniformly convergent series satisfy the identity

$$G_a(x) = G_b(x) + (a-b)\sum_{j=1}^{\infty} a^{j-1}(-1)^j G_b(T_{3^j}(x)). \tag{12}$$

Indeed, after expansion, the coefficient of T_{3^k} on the right is

$$(-1)^k \left[b^k + (a-b)\sum_{j=1}^{k} a^{j-1}b^{k-j} \right] = (-a)^k.$$

Rearrangement is justified by absolute uniform convergence.

Every coefficient multiplying a function on the right of (12) is positive. Each such function has norm B_b and value B_b at u, because $T_3(u) = -u$ and G_b is odd. Thus

$$G_a(u) = B_b\left(1 + \frac{a-b}{1-a}\right) = \frac{u}{1-a}.$$

All deficits from this maximum are nonnegative. Keeping the first deficit and using (11) proves $B - G_a(y) \geq (y-u)^2/2$. Oddness proves $B + G_a(y) \geq (y+u)^2/2$. Their minimum is (10), and also proves uniqueness of the two extrema. □

For later use, $P_0 G = 0$ by oddness. Lemma 7.1 applied to $G = \mathcal{S}T_1$ therefore gives

$$P_1 G = P_2 G = T_1. \tag{13}$$

It follows that $G \circ T_d$ has initial best approximant zero for $0 \leq n < d$, and best approximant T_d for $d \leq n < 3d$. The residuals in these two ranges are $G \circ T_d$ and $-aG \circ T_{3d}$.

9 From finite contact inequalities to the whole interval

The point of the larger contraction is that the baseline gap and an arbitrary polynomial-forcing perturbation can be compared at the same scale. No stationarity conditions on the perturbation are needed.

Lemma 9.1 (A gap at the exit scale)**.** *Let $0 < |s| \leq \delta := 1/16$, and choose $J \geq 0$ minimal with $3^J|s| \geq \delta$. Then*

$$B - G(\cos(\pi/4 + s)) \geq \frac{a^J}{4608}. \tag{14}$$

Proof. Minimality gives $\delta \leq 3^J|s| \leq 3\delta$. The exact phase relation, using $3^J \equiv 1$ or $3 \pmod 8$ according to the parity of J, is

$$(-1)^J G_b(\cos(3^J(\pi/4 + s))) = G_b(\cos(\pi/4 + (-1)^J 3^J s)).$$

On $[\pi/4 - 3\delta, \pi/4 + 3\delta]$, sine is at least $1/2$. Thus the argument of the final G_b is at distance at least $\delta/2$ from u, and its deficit is at least $\delta^2/8 = 1/2048$ by (11). For $J \geq 1$ the coefficient of this term in (12) is $(a-b)a^{J-1} = (4/9)a^J$. For $J = 0$ the coefficient is $1 \geq 4/9$. Discarding all other nonnegative deficits proves (14). □

For a polynomial $w = \sum w_k T_k$, the bounds

$$\|w\|_\infty \le \sum |w_k|, \qquad \sup_\theta \left| \frac{d}{d\theta} w(\cos\theta) \right| \le \sum k|w_k|$$

follow directly from the cosine representation. The second bound is in the angular variable, not the spatial variable.

Lemma 9.2 (Perturbation oscillation)**.** *Let $H = \mathcal{S}h$ and $D = H - v$, where h, v are polynomials. Suppose*

$$\|h\|_\infty \le M, \quad \sup_\theta |(h(\cos\theta))'| \le L, \quad \|v\|_\infty \le V, \quad \sup_\theta |(v(\cos\theta))'| \le L_v.$$

Let $d \ge 1$ and let J be a nonnegative integer with $\delta \le 3^J d|t| \le 3\delta$. Then

$$|D(\cos(\theta + t)) - D(\cos\theta)| \le C_d a^J, \qquad C_d = 5M + \frac{15L}{64d} + \frac{3L_v}{16d}. \tag{15}$$

Also $\|D\|_\infty \le K := \frac{5}{2}M + V$.

Proof. The jth forcing difference is at most $\min\{L3^j|t|, 2M\}$. Splitting the series before J gives

$$\begin{aligned} |H(\cos(\theta + t)) - H(\cos\theta)| &\le L|t| \sum_{j<J} (3a)^j + 2M \sum_{j \ge J} a^j \\ &\le \left(\frac{15L}{64d} + 5M \right) a^J. \end{aligned}$$

Here $3a - 1 = 4/5$ and $3^J|t| \le 3/(16d)$. Moreover $3^{-J} \le a^J$, so $L_v|t| \le 3L_v a^J/(16d)$. This proves (15). The norm bound follows from $\|H\|_\infty \le M/(1-a) = 5M/2$. □

Proposition 9.3 (Quantitative finite-contact stability)**.** *Let*

$$R(x) = sAG(T_d(x)), \qquad A > 0, \quad s \in \{-1, 1\}, \quad d \ge 1.$$

Its extremal points are the $2d$ zeros of T_{2d}, namely $x_i = \cos((2i+1)\pi/(4d))$, $0 \le i < 2d$. Put $\sigma_i = R(x_i)/(AB)$. For $D = \mathcal{S}h - v$, suppose

$$\sigma_i D(x_i) \le \lambda \qquad (0 \le i < 2d). \tag{16}$$

With the constants of Lemma 9.2, set

$$\eta_0 = \frac{1}{1296}, \qquad \gamma = \frac{1}{8 \cdot 1296^2}.$$

If $\varepsilon > 0$ satisfies

$$\varepsilon C_d \le \frac{A}{4608}, \qquad \varepsilon(K + |\lambda|) \le A\gamma, \tag{17}$$

then

$$|R(x) + \varepsilon D(x)| \le AB + \varepsilon\lambda \qquad (x \in [-1, 1]). \tag{18}$$

Every equality in (16) *is an equality in* (18) *with sign σ_i.*

Proof. Use $x = \cos\theta$. Around every extremal angle θ_i take the neighborhood $|\theta - \theta_i| \le \eta_0/d$. By periodicity, reflection, and oddness, the signed baseline value reduces to $AG(\cos(\pi/4 + s'))$, with $|s'| = d|\theta - \theta_i|$. For $s' \neq 0$ choose its exit scale J. Lemmas 9.1 and 9.2 give

$$\sigma_i(R(x) + \varepsilon D(x)) \le AB - \frac{Aa^J}{4608} + \varepsilon\lambda + \varepsilon C_d a^J \le AB + \varepsilon\lambda.$$

At the contact itself this follows from (16).

We must also control the opposite sign. The neighborhood radius ensures $J \ge 4$. Apply the forcing estimate in Lemma 9.2 to $G = \mathcal{S}T_1$, taking $d = 1$, $M = L = 1$, and omitting v. Then

$$|G(\cos(\pi/4 + s')) - B| \le \frac{335}{64}a^J \le \frac{335}{64}\left(\frac{3}{5}\right)^4 < \frac{679}{1000} < \frac{B}{2}.$$

Thus $\sigma_i R(x) > AB/2$. The second inequality in (17) gives $\varepsilon \|D\|_\infty \le A\gamma < AB/2$. The perturbed residual retains this sign, proving the absolute bound throughout the neighborhoods. At $s' = 0$, sign retention follows directly from $AB - \varepsilon K > 0$.

Outside these neighborhoods, the angular distance of $d\theta$ from every preimage of $\pm u$ is at least η_0. Monotonicity of cosine between its extrema, and the sine bound $1/2$ in a short interval adjacent to each preimage, give $\operatorname{dist}(\cos(d\theta), \{-u, u\}) \ge \eta_0/2$. By (10), $|R(x)| \le AB - A\gamma$. Therefore

$$|R(x) + \varepsilon D(x)| \le AB - A\gamma + \varepsilon K \le AB + \varepsilon\lambda.$$

Finally, $AB + \varepsilon\lambda > 0$ because $\varepsilon|\lambda| \le A\gamma < AB$. At an active contact the signed value is exactly $AB + \varepsilon\lambda$, as claimed. □

In both constructions below the exact certificates give

$$M < 32, \quad L < 128, \quad V < 32, \quad L_v < 128, \quad |\lambda| < 32. \tag{19}$$

All residuals have $A \ge 3/5$ and $d \ge 3$. Thus $C_d < 178$ and $K + |\lambda| < 144$. The single explicit choice

$$\varepsilon = 10^{-10} \tag{20}$$

satisfies (17), since

$$\frac{178}{10^{10}} < \frac{3}{5 \cdot 4608}, \qquad \frac{144}{10^{10}} < \frac{3}{5 \cdot 8 \cdot 1296^2}.$$

Consequently, any of the finite certificates below with the stated bounds is automatically a certificate on the whole interval.

10 The normalized oddness counterexample

Define

$$h_{\mathrm{o}} = (\eta - \sqrt{3} - 1)T_0 - \sqrt{3}\,T_2 + T_4 - 3T_5 + \eta T_6 + \beta T_7, \tag{21}$$

$$\eta = \frac{16\cos(\pi/18) - 26\cos(\pi/6) - 26\cos(5\pi/18)}{15}. \tag{22}$$

To specify β exactly, put $c_j = \cos(j\pi/36)$ and set

$$\begin{aligned} 7132971\beta = {} & -21647763 + 7264892c_1 - 16364700c_2 + 2333622c_3 - 20710080c_4 \\ & + 7960924c_5 - 8377290c_6 + 6354480c_7 - 6380100c_8 \\ & + 8032814c_9 - 667710c_{10} + 24678c_{11}. \end{aligned} \tag{23}$$

Numerically $\eta \approx -1.564814286803340$ and $\beta \approx -5.914876184323304$. These decimals are descriptive only; (22)–(23) define the coefficients.

Set

$$f_{\mathrm{o}} = G \circ T_3 + \varepsilon \mathcal{S} h_{\mathrm{o}} = \mathcal{S}(T_3 + \varepsilon h_{\mathrm{o}}). \tag{24}$$

We next give the three finite certificates needed for this function.

10.1 Exact definition of the correction polynomials

Let

$$x_j = -\cos\frac{(2j+1)\pi}{36}, \quad 0 \le j < 18, \qquad \sigma_j = (1,-1,-1,1)_{j \bmod 4}.$$

The coarse grid is $y_i = x_{3i+1}$, $0 \le i < 6$, with signs $\tau = (-1,1,1,-1,-1,1)$. These are respectively the extrema of $-aG \circ T_9$ and $G \circ T_3$.

Define the explicit grid functions

$$W_k(x_j) = T_k(x_j) - aT_{3k}(x_j) + \frac{a^2}{1+(-1)^k a} T_{9k}(x_j).$$

They equal $(\mathcal{S}T_k)(x_j)$: after two iterates the finite orbit is on the two-cycle $\{\pm u\}$, and the remaining series is geometric. Equivalently, the exact identities $W_k(x_j) + aW_k(T_3 x_j) = T_k(x_j)$ determine these values uniquely. Write $H_j = \sum_{k=0}^{7} (h_{\mathrm{o}})_k W_k(x_j)$.

The correction polynomials v_1, v_3, v_5 and scalars $\lambda_2, \lambda_4, \lambda_6$ are specified by the contact equations

$$\begin{aligned}
v_1(y_i) + \tau_i \lambda_2 &= H_{3i+1}, && i \in I_2 := \{0,1,4,5\},\\
v_3(x_j) + \sigma_j \lambda_4 &= H_j, && j \in I_4 := \{0,2,4,13,15,17\},\\
v_5(x_j) + \sigma_j \lambda_6 &= H_j, && j \in I_6 := \{0,2,4,6,11,13,16,17\},
\end{aligned}$$

with degrees at most $1, 3, 5$, respectively. The systems are consistent for the exact values (22)–(23). Each has full column rank, so the correction and scalar are unique. Their exact rational coordinates in the field described below are included in `odd_certificate.json`.

For orientation, their Chebyshev coefficients are:

coefficient	v_1	v_3	v_5
T_0	-3.310540684	-3.310540684	-3.380936697
T_1	-0.781038720	0.805215314	-1.373864474
T_2	0	-1.138509891	-0.999856812
T_3	0	-2.959735280	1.640309342
T_4	0	0	-0.070396013
T_5	0	0	-4.379580520

The corresponding scalars are approximately 4.600544015619, 7.815043886301, and 6.118310129767.

Lemma 10.1 (Odd-pattern finite certificate)**.** *The above corrections satisfy the bounds* (19)*. At every coarse or fine node, as appropriate,*

$$\tau_i(H_{3i+1} - v_1(y_i)) \le \lambda_2, \qquad \sigma_j(H_j - v_{n-1}(x_j)) \le \lambda_n \quad (n = 4, 6).$$

Equality holds exactly on I_2, I_4, I_6, respectively. Each active set has alternating signs. Every inactive slack is greater than $3/50$.

Exact arithmetic verification. Put $z = 2\cos(\pi/36)$. All entries lie in

$$\mathbb{Q}[z], \qquad M(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1 = 0.$$

The relevant root is isolated in $(1.98, 2)$. To see uniqueness there, write $M = S_6^2 - 3$, where $S_6 = z^6 - 6z^4 + 9z^2 - 2$; on that interval $S_6 > 0$ and $S_6' = 6z(z^2-1)(z^2-3) > 0$.

The program `verify_odd.py` constructs every coefficient by exact field arithmetic, checks the contact equations, and checks the displayed closed expressions for η, β. It encloses the root by 250 rational bisections and evaluates every nonzero slack by rational interval arithmetic. The lower bounds exceed $3/50$; the least positive slacks for the three certificates exceed 1.5265, 1.6678, and 0.0619, respectively. Exact interval bounds also establish (19), $\eta < 0$, and positivity of the three scalars.

The independently implemented `audit_odd.py` starts from the serialized rational coefficients. It checks the root isolation, ordered nodes, cubic node map, finite functional equation for the actual H, all slack identities, their signs, and the alternating active sets. It also checks (17) directly for $\varepsilon = 10^{-10}$. These programs use rational arithmetic for every acceptance condition; floating-point conversion is used only in printed summaries. □

10.2 Completion of the first construction

For degrees $1, 2$, the baseline approximant is zero, so the residual for the candidate εv_1 is $G \circ T_3 + \varepsilon(\mathcal{S}h_\mathrm{o} - v_1)$. For degrees $3, 4$ and $5, 6$, the baseline approximant is T_3, and the residual is $-aG \circ T_9 + \varepsilon(\mathcal{S}h_\mathrm{o} - v)$. Apply Proposition 9.3 with $(A, s, d) = (1, 1, 3)$ or $(a, -1, 9)$. The finite certificate provides respectively four, six, and eight alternating contacts, while the candidates have degrees at most $1, 3, 5$. Chebyshev alternation therefore gives

n	$P_n f_\mathrm{o}$	$E_n(f_\mathrm{o})$
$1, 2$	εv_1	$B + \varepsilon\lambda_2$
$3, 4$	$T_3 + \varepsilon v_3$	$aB + \varepsilon\lambda_4$
$5, 6$	$T_3 + \varepsilon v_5$	$aB + \varepsilon\lambda_6$.

Proposition 7.2 now proves every odd/even plateau.

The identity $h_\mathrm{o}(0) = 0$ follows directly from $T_{2j}(0) = (-1)^j$. Thus (7) at zero gives $f_\mathrm{o}(0) = 0$. At the endpoints,

$$h_\mathrm{o}(1) + h_\mathrm{o}(-1) = 4(\eta - \sqrt{3}).$$

Since $T_3(\pm 1) = \pm 1$ and $G \circ T_3$ is odd,

$$f_\mathrm{o}(1) + f_\mathrm{o}(-1) = \frac{4\varepsilon(\eta - \sqrt{3})}{1 + a} = \frac{5}{2}\varepsilon(\eta - \sqrt{3}) \neq 0. \tag{25}$$

This proves the first assertion of Theorem 6.1.

11 The evenness counterexample

This construction uses the baseline $G \circ T_6$ and a degree-eight forcing perturbation. We specify its coefficients by a small, explicit linear system, avoiding long rational coordinate lists in the text.

11.1 Exact algebraic definition

The unknown forcing is

$$h_{\mathrm{e}} = c_1T_1 + c_2T_2 + c_3T_3 + c_4T_4 + c_5T_5 + c_7T_7 + c_8T_8.$$

Thus $c_0 = c_6 = 0$. Introduce correction polynomials w_0, w_2, w_4, w_6 of degrees at most $0, 2, 4, 6$, and scalars $\mu_1, \mu_3, \mu_5, \mu_7$.

For $m = 12, 36$, define the increasing nodes

$$x_j^{(m)} = -\cos\frac{(2j+1)\pi}{2m}, \qquad 0 \le j < m.$$

Use $m_n = 12$ for $n = 1, 3, 5$ and $m_7 = 36$. Let

$$\sigma_{n,j} = \begin{cases} (1,-1,-1,1)_{j \bmod 4}, & n = 1, 3, 5, \\ (-1,1,1,-1)_{j \bmod 4}, & n = 7. \end{cases}$$

For a grid angle θ set $J = 1$ when $m = 12$ and $J = 2$ when $m = 36$, and define

$$W_k^{(m)}(\cos\theta) = \sum_{i=0}^{J-1}(-a)^i\cos(k3^i\theta) + \frac{\sum_{i=J}^{J+3}(-a)^i\cos(k3^i\theta)}{1-a^4}. \tag{26}$$

After J iterates these node orbits have period dividing four. Therefore (26) is the exact value of $\mathcal{S}T_k$ at the node.

Write $\mathcal{H}_j^{(m)} = \sum_k c_k W_k^{(m)}(x_j^{(m)})$. The defining equations are

$$w_{n-1}(x_j^{(m_n)}) + \sigma_{n,j}\mu_n = \mathcal{H}_j^{(m_n)}, \qquad j \in J_n, \tag{27}$$

where

$$\begin{aligned} J_1 &= \{3, 6, 8\}, \\ J_3 &= \{1, 3, 4, 6, 8, 10\}, \\ J_5 &= \{0, 2, 4, 6, 7, 9, 11\}, \\ J_7 &= \{3, 6, 8, 13, 19, 22, 27, 30, 32\}. \end{aligned}$$

Add the two normalizations

$$[T_2]w_2 = 1, \qquad [T_4]w_4 = -1. \tag{28}$$

Here $[T_k]w$ denotes the coefficient in the Chebyshev basis. There are 27 equations for the $7 + 1 + 3 + 5 + 7 + 4 = 27$ unknowns. Exact elimination shows that this system is nonsingular. Equations (26)–(28) therefore define the forcing and corrections unambiguously by real algebraic numbers.

Their approximate forcing coefficients are

k	c_k
1	−0.017561786853260
2	0.286725254171375
3	0.018613141564351
4	−0.862027229474975
5	−0.013885677377745
7	0.026299792766218
8	−0.253316709984462.

The exact rational coordinate lists are in `even_certificate.json`. Define

$$f_{\mathrm{e}} = G \circ T_6 + \varepsilon\mathcal{S}h_{\mathrm{e}} = \mathcal{S}(T_6 + \varepsilon h_{\mathrm{e}}). \tag{29}$$

Lemma 11.1 (Even-pattern finite certificate)**.** *For each $n = 1, 3, 5, 7$ and every point of the corresponding grid,*

$$\sigma_{n,j}\big(\mathcal{H}_j^{(m_n)} - w_{n-1}(x_j^{(m_n)})\big) \le \mu_n.$$

The equality sets are exactly J_n. Alternating subsets of cardinality $n+2$ are J_1, J_5, J_7, and $\{1, 3, 6, 8, 10\} \subset J_3$. All nonzero slacks exceed $1/50$. The coefficient bounds (19) *hold, and*

$$c_1 + c_3 + c_5 + c_7 > \frac{1}{100}. \tag{30}$$

Exact arithmetic verification. All coefficients belong to the real algebraic field generated by $z = 2\cos(\pi/72)$, whose defining polynomial is

$$\begin{aligned} M_{24}(z) = {} & z^{24} - 24z^{22} + 252z^{20} - 1520z^{18} + 5814z^{16} - 14688z^{14} \\ & + 24752z^{12} - 27456z^{10} + 19305z^8 - 8008z^6 + 1716z^4 - 144z^2 + 1. \end{aligned}$$

It satisfies $M_{24}(z) = M(z^2 - 2)$, where M is the polynomial used in the odd certificate. The relevant root is isolated in $(1.998, 2)$; the same monotonicity argument, applied to $z^2 - 2$, proves uniqueness.

The generator `generate_even.py` solves the 27-by-27 system by exact arithmetic, with a nonzero pivot at every step. It checks every functional identity, equality, and slack and encloses the root by 400 rational bisections. The minimum positive slacks for $n = 1, 3, 5, 7$ exceed 0.1336, 0.0936, 0.0965, and 0.0230, respectively. Exact coefficient bounds give $M < 2$, $L < 34$, $V < 2$, $L_v < 72$, and $|\mu_n| < 1$, which imply (19); it also verifies (30).

The independent standard-library program `audit_even.py` uses a separate implementation of rational polynomial arithmetic. From the serialized coefficients it reconstructs all 36 fine nodes, verifies their ordering and cubic map, detects the finite orbit cycles, and evaluates the exact orbit sums. It then verifies every slack, equality, alternating set, coefficient bound, and (17). No floating-point tolerance is used to accept any identity or inequality. □

For reference, the scalar error corrections are approximately

$$\begin{aligned} \mu_1 &= 0.577201789619347, & \mu_3 &= 0.223648399026073, \\ \mu_5 &= -0.129582723095644, & \mu_7 &= 0.276968241048616. \end{aligned}$$

The negative value of μ_5 causes no difficulty: the stability proposition uses $|\mu_5|$ in its smallness condition, and the total error remains positive.

11.2 Completion of the second construction

By (13) and Chebyshev composition, the baseline $G \circ T_6$ has best approximant zero for degrees 0 through 5 and best approximant T_6 for degrees $6, 7$. Its residuals are consequently $G \circ T_6$ and $-aG \circ T_{18}$.

Apply Proposition 9.3 with $(A, s, d) = (1, 1, 6)$ for the first three certificates and $(a, -1, 18)$ for the fourth. Alternation gives

n	$P_n f_{\mathrm{e}}$	$E_n(f_{\mathrm{e}})$
$0, 1$	εw_0	$B + \varepsilon\mu_1$
$2, 3$	εw_2	$B + \varepsilon\mu_3$
$4, 5$	εw_4	$B + \varepsilon\mu_5$
$6, 7$	$T_6 + \varepsilon w_6$	$aB + \varepsilon\mu_7$.

Since $\deg(T_6 + \varepsilon h_{\mathrm{e}}) \le 8$, Proposition 7.2 extends these equalities to every even/odd pair.

Finally, $G \circ T_6$ is even, while the endpoint difference of the perturbation is

$$f_{\mathrm{e}}(1) - f_{\mathrm{e}}(-1) = \frac{2\varepsilon}{1+a}(c_1 + c_3 + c_5 + c_7) > \frac{\varepsilon}{80} > 0. \tag{31}$$

This proves the second assertion of Theorem 6.1.

12 Reproducibility and formal proof status

The source and compiled version of this complete manuscript are `paper/lorentz_counterexamples.tex` and `paper/lorentz_counterexamples.pdf`. From the repository root, the independent exact checks are

```
python3 scripts/check_certificate.py
python3 certificates/plateau/audit_odd.py
python3 certificates/plateau/audit_even.py
```

The Lean project is in `lean/`. All three counterexample existence theorems are now proved with only Lean's standard logical axioms. The two plateau proofs include exact algebraic root brackets, invariant grid recurrences, all signed slacks, completeness and simplicity of the contact sets, analytic persistence, and the induction to every degree. The independent statements in `lean/Challenge.lean` match the proved declarations in `lean/Solution.lean`. None of the proved dependencies contains a proof hole or a custom mathematical axiom. The project builds with Lean and Mathlib `v4.35.0-rc2`, the Palomar-supported toolchain used for this preparation. These status distinctions do not replace any hypothesis in the mathematical proofs above.

The common-origin proof in Part I corrects the strict Lipschitz inequality at coincident points in the earlier standalone PDF: the valid inequality is non-strict. Its explicit perturbation range is proved in the manuscript; all three Lean endpoints are existential in the positive perturbation size. The plateau arguments in Part II justify the fixed value 10^{-10} by their stated analytic estimates and exact coefficient checks; this numerical choice is not part of the Lean endpoint claims.

The original supplied manuscripts and certificate package are retained under `archive/`. The current standalone certificate scripts live under `certificates/plateau/`; references to those script basenames in Part II refer to that directory.

References

[1] E. B. Saff and R. S. Varga, *Remarks on some conjectures of G. G. Lorentz*, Journal of Approximation Theory **30** (1980), 29–36. `https://www.math.kent.edu/~varga/pub/paper_126.pdf`.